

Not a slide deck.

A build specification.

Instructions + figures for Claude Design

A Transformer on a \$5 microcontroller

Event label is unresolved:
confirm before deck generation

What this PDF gives Claude

01 A fixed 21-slide narrative

Every slide has one judge takeaway and exact minimal copy.

02 Reusable scientific figures

Eleven charts are supplied as editable SVG and high-resolution PNG.

03 Measured-versus-derived discipline

Labels prevent projections, estimates, and hardware results from being conflated.

04 Photo insertion instructions

Four named slots are reserved for the user's real device and bench photographs.

OUTPUT REQUEST

Create a 16:9 presentation from this specification. Do not redesign this PDF itself.

Use supplied figures and replace only the marked photo assets.

Prompt Claude Design with this contract

Create a 21-slide, 16:9 technical competition presentation titled 'A Transformer on a \$5 microcontroller'. The audience is an expert HPC jury that already understands attention, quantization, MFU, and parallel scaling. Use the slide specifications on pages 05–25 exactly: preserve the proposed slide order, exact numerical values, units, and measured/derived/projected labels. Keep on-slide text sparse: one conclusion headline, at most three short support statements, and one dominant evidence object. Use the supplied SVG figures without turning them into decorative dashboards. Use a restrained scientific visual system: off-white or deep-navy fields, high-contrast typography, one orange difficulty accent, one blue systems accent, and one green verified-result accent. Insert only user-supplied photographs in the four named photo slots; do not synthesize boards or laboratory scenes. Never imply this is a compliant GPU kernel, never present a non-run case as a benchmark, never turn correctness equivalence into task-model accuracy, and keep host-transfer exclusions visible. Add short source notes in 8–10 pt type. Return an editable deck and preserve chart axes.

DELIVERY CHECKLIST

☐ 21 slides, 16:9 ☐ editable vector figures ☐ four real-photo slots ☐ no invented results ☐ all axes and units retained ☐ measured / derived / projected labels visible

Competition concept, rules, and evaluation

These facts control the story; they are not optional design suggestions.

TASK

Optimize a fixed pre-LayerNorm Transformer body

Fourteen official (B,S,D,H,F,L) shapes. Output remains [B,S,D]. Final LayerNorm is included.

Accuracy failure blocks performance scoring.

ELEMENTWISE GATE

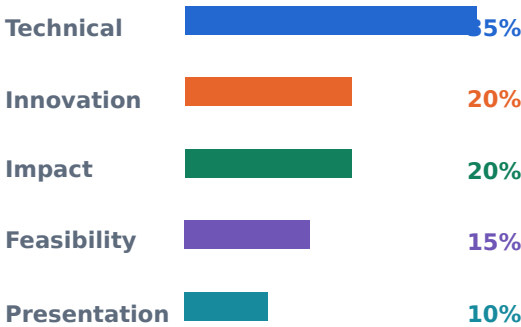
$$|user - ref| \leq 0.002$$

OR

$$|user - ref| \leq 0.02 |ref|$$

Official evaluation: 5 trials. Project validation uses stronger seeded gates where reported.

JUDGING WEIGHTS



FRAMING THAT MUST STAY HONEST

1. This project preserves the model semantics and numerical gate, but it is not a GPU-kernel submission lane.
2. Performance means complete forward time on physical ESP32-C3 hardware unless the label says host-only, estimate, or projection.
3. The implementation uses seeded random weights for equivalence testing; do not claim trained-model application accuracy.

Make it look like an HPC result, not an AI template

HIERARCHY

- Conclusion headline: 32–40 pt
- Support copy: 18–22 pt
- Source/caveat: 8–10 pt
- ≤35 on-slide words, excluding labels
- One visual argument per slide

FIGURES

- Use supplied SVG first; PNG is fallback
- No 3D bars, gauges, or fake telemetry
- Keep axes, units, and log-scale labels
- Direct-label important values
- Orange=difficulty; blue=system; green=pass

PHOTOGRAPHS

- photo_01_hero_device
- photo_02_board_closeup
- photo_03_cluster_wide
- photo_04_profiler_bench
- Crop documentary, not cinematic

ASSET BUNDLE

figure_assets/fig01...fig11 SVG + 240 dpi PNG

optimization_ladder.csv

benchmark_cases.csv

Do not trace charts from PDF screenshots. Import the SVG or use the CSV to rebuild them with the same scale and labels.

Open with the physical claim

Judge takeaway: The hardware constraint is the hook; the result is real and measured.

EXACT ON-SLIDE COPY

A Transformer on a \$5 microcontroller

- ESP32-C3 · 160 MHz · no FPU · 321 KB usable SRAM
- [EVENT NAME — CONFIRM] · team / affiliation

21.1× one-board speedup

COMPOSITION INSTRUCTION

Use the user's strongest close-up device photograph as the dominant field. Put the title in clear negative space; keep only one result badge.

WIREFRAME ONLY

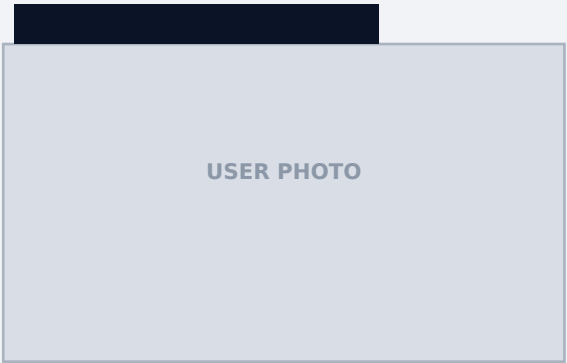


PHOTO · photo_01_hero_device

DATA / EVIDENCE

Baseline case 2: 42.15 s.
Optimization-log endpoint: 1.996 s;
authoritative repeat median: 1.990 s.

MUST PRESERVE

Do not use a synthetic chip image. Do not imply the \$5 price is a formal bill of materials.

SOURCE PATH

docs/report/index.html
benchmarks/README.md

State the deployment vision without overselling it

Judge takeaway: Local inference on tiny endpoints is valuable because connectivity, power, and hardware budgets are finite.

EXACT ON-SLIDE COPY

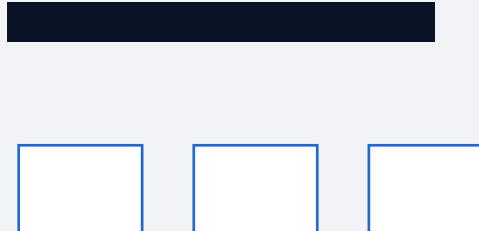
What if every sensor could run its own Transformer?

- Local inference: private, resilient, low-latency
- Constraint: consumer- and industrial-grade accelerators do not fit every endpoint
- Potential: offline industrial sensing and private local monitoring

COMPOSITION INSTRUCTION

Use a three-node edge-to-cloud schematic: sensor → local decision → optional cloud. Make the local path solid and the cloud path secondary/dashed.

WIREFRAME ONLY



NO EXTERNAL ASSET

DATA / EVIDENCE

This is motivation, not a demonstrated application workload. The benchmark uses seeded random weights and a fixed Transformer body.

MUST PRESERVE

Use the words 'potential' or 'vision'. Do not claim energy, privacy, or latency measurements that are absent.

SOURCE PATH

COMPETITION_RULES.MD
docs/report/index.html

Define exactly what the project proves

Judge takeaway: The contribution is systems co-design under extreme constraints, not a smaller model architecture.

EXACT ON-SLIDE COPY

We kept the Transformer. We changed the execution system.

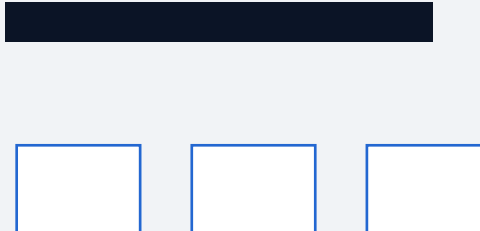
- Fixed-point attention and integer residuals
- SRAM-lifetime scheduling and register-tiled RV32 kernels
- Two distributed execution modes across eight boards

10 / 14 official shapes measured

COMPOSITION INSTRUCTION

Use three equal blocks—Numerics, Memory, Parallelism—feeding one verified-output block. This is the map for the rest of the talk.

WIREFRAME ONLY



NO EXTERNAL ASSET

DATA / EVIDENCE

All reported hardware results preserve the elementwise competition gate. Four official shapes are explicitly not device runs.

MUST PRESERVE

Say 'Transformer body' rather than foundation model or LLM. No trained-model claims.

SOURCE PATH

COMPETITION_RULES.MD
docs/report/index.html

Introduce the exact constraint surface

Judge takeaway: The ESP32-C3 is a scalar RISC-V microcontroller with neither accelerator nor FPU.

EXACT ON-SLIDE COPY

ESP32-C3: a deliberately hostile Transformer target

- RV32IMC · single core · 160 MHz
- No floating-point unit
- 321,296 B usable DRAM segment · 4 MB flash · 3.1 MB app partition

COMPOSITION INSTRUCTION

Place the user's board close-up on the left. On the right, use four technical callouts anchored to the real components; avoid lifestyle-card styling.

WIREFRAME ONLY

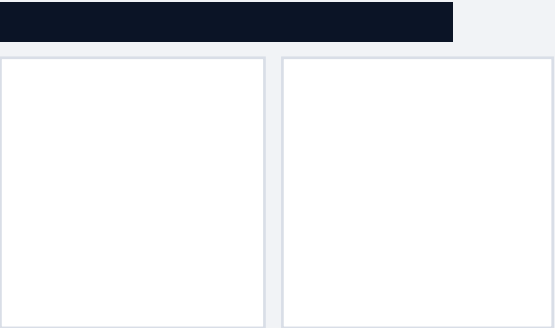


PHOTO · photo_02_board_closeup

DATA / EVIDENCE

Usable SRAM is the linked dram0_0_seg, not the marketing SRAM total. Board: Seeed XIAO ESP32-C3.

MUST PRESERVE

If comparing to ESP32-S3, state that S3 has an FPU and larger memory options; do not imply it is the tested device.

SOURCE PATH

docs/
esp32_implementation_summary.md
docs/HARDWARE_COMPARISON.md

Explain why no FPU changes the algorithm

Judge takeaway: Every fp32 inner-loop operation becomes software work; fixed-point is an architectural requirement.

EXACT ON-SLIDE COPY

Without an FPU, fp32 becomes a function call

- C3 fp32: add 100 cycles · divide 102 · cosf 2,377
- S3 hardware FPU: 25 · 69 · 121 cycles
- Baseline attention consumed 30.09 s of a 42.15 s forward

COMPOSITION INSTRUCTION

Use the log-scale C3-versus-S3 chart as the dominant object. Add one small annotation connecting software fp32 to attention's baseline wall time.



ASSET PREVIEW · fig01 fpu cost c3 vs s3.sva



DATA / EVIDENCE

Cycle figures are documented Espressif measurements quoted in the repository. The C3 has RV32IMC, not the F extension.

MUST PRESERVE

Q15 multiplication uses a native integer instruction; do not claim a universal one-cycle latency.

SOURCE PATH

benchmarks/case-02/optimisation/
esp32-baseline/optimisations/
research.md
docs/
esp32_fastest_kernels_research.md

Make the memory scale physically legible

Judge takeaway: A single tensor can exceed the entire machine; algorithmic tiling is mandatory.

EXACT ON-SLIDE COPY

321 KB is not a memory budget. It is a scheduling constraint.

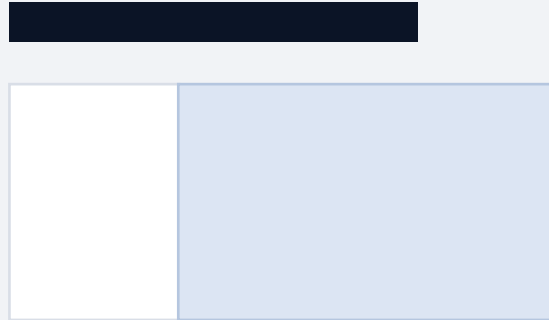
- Final case-2 arena: 274,340 B used · 46,956 B free
- Case 13 one activation: 524 KB
- Case 14 full-batch KV state: 26.2 GB

COMPOSITION INSTRUCTION

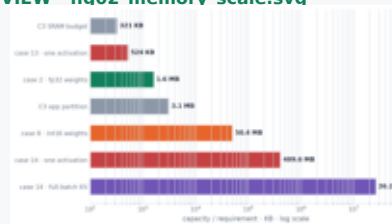
Use the log-scale memory figure as the main visual and the final SRAM stack as a small inset.

Keep budget bars visually distinct from requirements.

WIREFRAME ONLY



ASSET PREVIEW · fig02 memory scale.svg



DATA / EVIDENCE

Secondary inset:
figure_assets/fig03_final_sram_budget.svg.
The earlier 384 B-free build is a different intermediate state and must not be mixed in.

MUST PRESERVE

Always show the logarithmic axis label. Distinguish SRAM, flash/app partition, weights, and activations.

SOURCE PATH

docs/report/index.html
benchmarks/case-02/optimisation/
esp32-baseline/optimisations/
25_tinyprof.md

Show that profiling—not intuition—selected the work

Judge takeaway: Attention was the measured bottleneck before optimization.

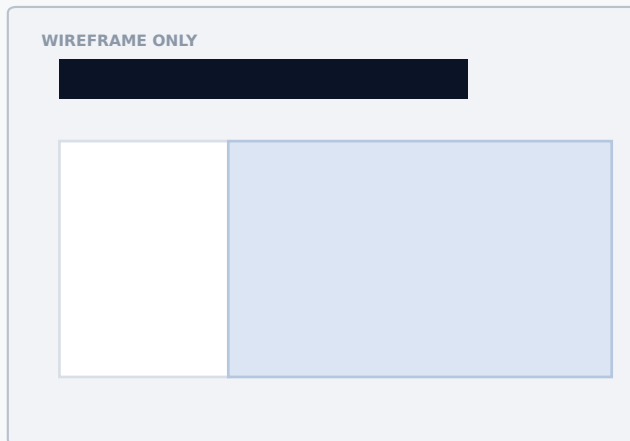
EXACT ON-SLIDE COPY

The baseline profile told us where not to guess

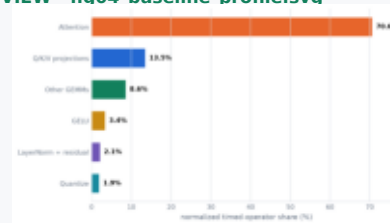
- 42.15 s complete case-2 forward
- 30.09 s attention wall time · 71.4%
- 512k expf calls plus fp32 QK dequantization and PV accumulation

COMPOSITION INSTRUCTION

Use the horizontal profile chart. Keep the 70.6% normalized operator share in the chart and the 71.4% raw wall ratio in a separate annotation.



ASSET PREVIEW · fig04 baseline profile.sva



DATA / EVIDENCE

Normalized timed-zone shares sum to about 100%; raw wall ratio is 30.09/42.15. They are related but not interchangeable.

MUST PRESERVE

Label the normalization difference rather than rounding both to one number.

SOURCE PATH

benchmarks/case-02/optimisation/
esp32-baseline/optimisations/
00_baseline_profile.md
docs/report/index.html

Present the work as an evidence-backed sequence

Judge takeaway: The 21× speedup came from accumulated numerical, kernel, fusion, and assembly changes.

EXACT ON-SLIDE COPY

No silver bullet: 24 engineering steps

- 42.15 s → 1.996 s across 24 steps grouped into 21 recorded milestones
- Numerics → kernels → fusion → hand-written RV32 assembly
- MFU: 2.0% → 42.2% of the derived scalar peak

21.1× faster

COMPOSITION INSTRUCTION

Use the optimization ladder full-width. Direct-label only the start, major phase boundaries, and endpoint; do not annotate every point.

WIREFRAME ONLY



ASSET PREVIEW · fiq05 optimization ladder.svg



DATA / EVIDENCE

The graph is cumulative, not an ablation. The authoritative repeat median is 1.990 s; keep it as a note, not a second endpoint.

MUST PRESERVE

Keep the y-axis logarithmic and distinguish 1.996 optimization endpoint from 1.990 repeat median.

SOURCE PATH

docs/report/index.html
benchmarks/case-02/optimisation/
README.md

Explain fixed-point attention as scale factoring

Judge takeaway: Quantization moved high-volume QK and PV work to integer arithmetic while retaining a stable rowwise softmax.

EXACT ON-SLIDE COPY

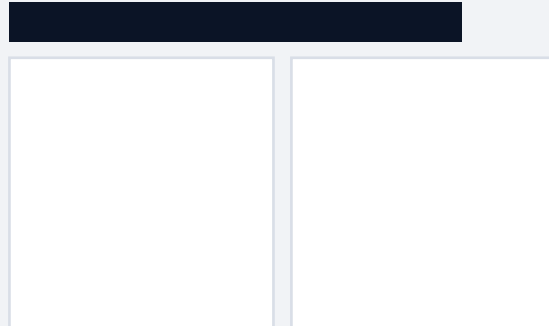
Move attention into fixed point—without moving the error gate

- QK: int64 Q15 dot products
- Softmax: LUT plus one fp32 scale per logit at the 13.70 s milestone
- PV: integer accumulation with fp32 only at row end; later made fully integer

COMPOSITION INSTRUCTION

Draw a before/after attention pipeline. Before: fp32 QK → expf → fp32 PV. After: Q15 QK → LUT softmax → Q15 PV. Put scale factors above arrows, not in prose.

WIREFRAME ONLY



NO EXTERNAL ASSET

DATA / EVIDENCE

Cumulative timing: 42.15 → 15.21 s after integer QK, then 13.70 s after first exp LUT. Integer PV lands later: 3.969 → 3.688 s.

MUST PRESERVE

Keep chronology exact. Do not describe the 13.70 s milestone as fully integer attention.

SOURCE PATH

benchmarks/case-02/optimisation/
README.md
docs/report/index.html

Make the GEMM result about registers, not generic tiling

Judge takeaway: A 4×2 accumulator tile balanced reuse and register pressure; larger tiles spilled.

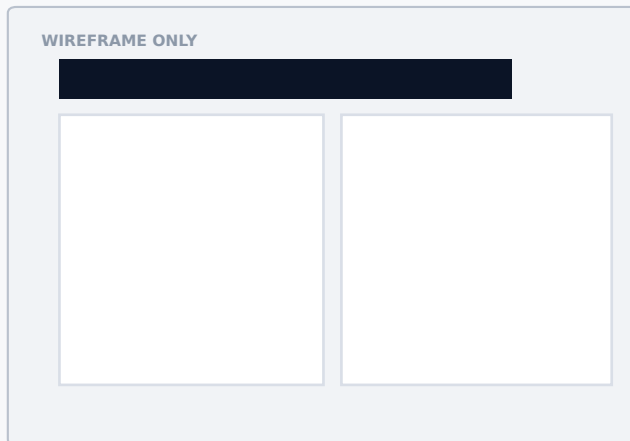
EXACT ON-SLIDE COPY

Register pressure—not tile area—set GEMM speed

- core4_v2: 7.48 cycles/MAC
- core5: 6.21 cycles/MAC
- ~20 live registers; 4×4 regressed from spills

COMPOSITION INSTRUCTION

Pair a simple 4×2 accumulator-grid diagram with the two-bar measured kernel comparison. Make the failed 4×4 attempt a small gray annotation.



ASSET PREVIEW · fiq06 gemm kernel.svg



DATA / EVIDENCE

Use the comparable core4_v2 → core5 microbenchmark pair. Other cumulative forward timings are not the same measurement.

MUST PRESERVE

Say cycles/MAC, not FLOP/s. Mention the unsuccessful larger tile to demonstrate engineering rigor.

SOURCE PATH

benchmarks/case-02/optimisation/
esp32-baseline/optimisations/
15_gemm_core5_iblk4.md

Connect fusion and integer residuals to eliminated passes

Judge takeaway: SRAM traffic fell because intermediate tensors stopped being materialized and rescanned.

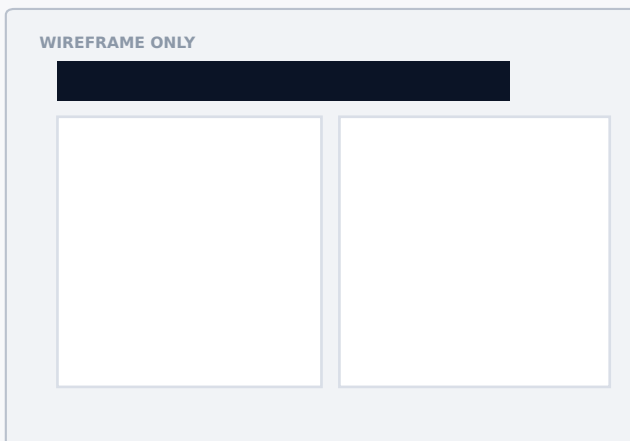
EXACT ON-SLIDE COPY

The fastest memory pass is the one the schedule deletes

- Fuse producer → scale bound → Q15 consumer
- Delete tensor write → amax scan → requantization passes
- Integer residual path: 268 → 10 instructions/element

COMPOSITION INSTRUCTION

Use a before/after dataflow with crossed-out write, amax, and requantize passes. Add the isolated timing deltas as a small evidence panel.



ASSET PREVIEW · fiq07 fusion deltas.svg



DATA / EVIDENCE

Isolated cumulative steps: 6.91→6.56 fused quant; 4.862→4.784 fused LN; 4.160→3.969 context fusion.

MUST PRESERVE

These deltas occur at different points in the cumulative sequence; do not add them as an independent total.

SOURCE PATH

benchmarks/case-02/optimisation/
README.md
docs/report/index.html

Put the numerical gate before the performance claim

Judge takeaway: Every reported speed result was accepted only after the elementwise equivalence test passed.

EXACT ON-SLIDE COPY

Speed counts only after every element passes

- Pass if absolute error ≤ 0.002 OR relative error $\leq 2\%$
- Case 2 device gate: 25/25 seeds
- Worst observed device absolute error across measured cases: $1.46e-3$

zero failing elements in reported gates

COMPOSITION INSTRUCTION

Use the tolerance bar plus the Boolean gate written once as a compact equation. Keep 'seeded random weights' in the footer.



ASSET PREVIEW · fig08_accuracy_gate.svg



DATA / EVIDENCE

Official rules specify 5 accuracy trials. The project reports stronger per-case gates; do not generalize 25 seeds to every row unless the table says so.

MUST PRESERVE

This is numerical equivalence, not task accuracy on a trained model.

SOURCE PATH

COMPETITION_RULES.MD
benchmarks/case-02/README.md
benchmarks/README.md

Use the real cluster as the narrative reset

Judge takeaway: After optimizing one core, the next limit was coordinated execution across eight memory-starved nodes.

EXACT ON-SLIDE COPY

Then we built an eight-node cluster

- Eight ESP32-C3 boards
- Two decompositions: batch data parallel + token-row split
- One shared problem: communication must coexist with the model arena

COMPOSITION INSTRUCTION

Make photo_03_cluster_wide nearly full bleed. Use only the headline and three small technical labels anchored to boards/router/wiring.

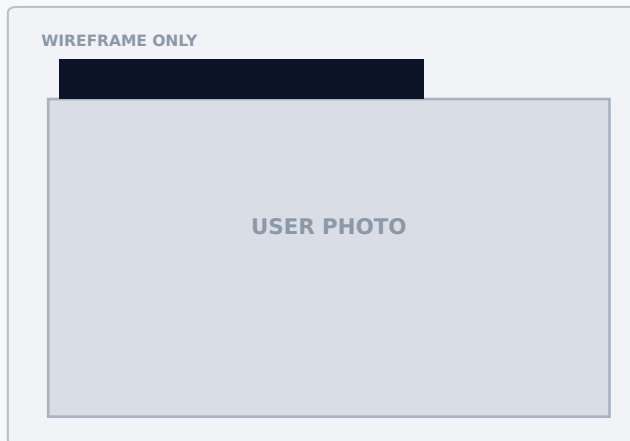


PHOTO · photo_03_cluster_wide

DATA / EVIDENCE

The cluster uses real WiFi/TCP/UDP paths described in the report; this is not a simulated topology.

MUST PRESERVE

Show the actual eight-board setup. No network-globe illustration.

SOURCE PATH

docs/MULTI_ESP32_DESIGN.md
docs/report/index.html

Explain how WiFi broke the memory budget

Judge takeaway: The team rewrote tensor lifetimes so networking and inference could fit on the same board.

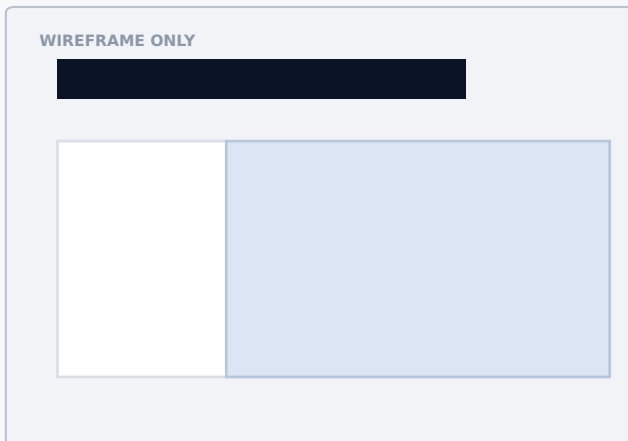
EXACT ON-SLIDE COPY

A node could compute—or communicate—but not both

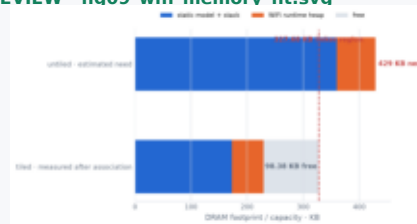
- Full forward + WiFi need: ~429 KB vs 327.68 KB linker region
- Rewrite: 16-row tiles + one attention head at a time
- Tiled + WiFi static: 173.06 KB · 98.38 KB heap after association

COMPOSITION INSTRUCTION

Use the three-bar memory fit chart and a small schedule strip showing row tiles/head-at-a-time. Make the ~100 KB deficit visible.



ASSET PREVIEW · fig09 wifi memory fit.sva



DATA / EVIDENCE

Tiling makes WiFi possible but costs performance: 4.215 s tiled versus 1.990 s best untiled single board.

MUST PRESERVE

Use 327.68 KB for this WiFi linker-region comparison; separately reported optimized map uses 321,296 B usable.

SOURCE PATH

benchmarks/case-02/optimisation/
esp32-baseline/optimisations/
24_sequence_tiled_wifi.md

Show why the final result is an overlap story

Judge takeaway: The team chose WiFi for fan-out and reduced exposed wait by streaming K/V per head.

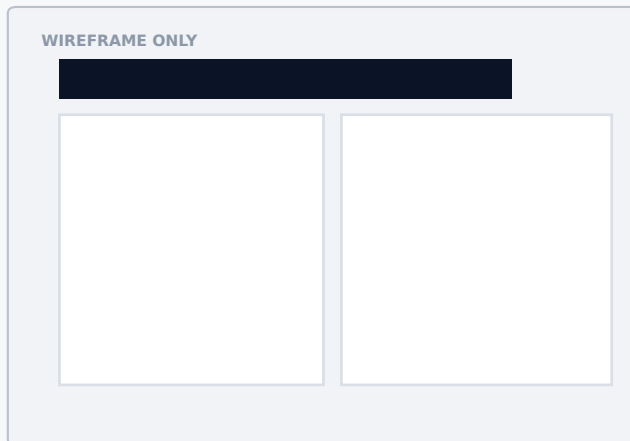
EXACT ON-SLIDE COPY

Bandwidth mattered. Overlap mattered more.

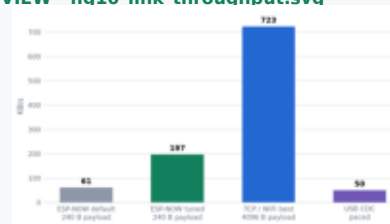
- TCP/WiFi link test peak: 723 KB/s
- First blocking TCP wait: 1.83 s
- Final UDP + NAK cluster wait: 5–88 ms

COMPOSITION INSTRUCTION

Use the throughput chart on the left and a projection/transfer/attention timeline on the right. Overlap the transfer bar with compute; do not imply zero communication.



ASSET PREVIEW · fig10 link throughput.svg



DATA / EVIDENCE

Why WiFi: one host reaches all eight nodes; only two usable host USB ports; paced USB CDC measured about 50 KB/s.

MUST PRESERVE

Throughput bars are a dedicated link benchmark. Wait-time numbers come from different implementation stages.

SOURCE PATH

benchmarks/case-02/multiboard/
esp32-linkbench/README.md
docs/WIFI_ON_A_COMPUTE_NODE.md

Teach the two decompositions with one comparison

Judge takeaway: Data parallelism is communication-free during the forward, while B=1 requires causal token-row work sharing.

EXACT ON-SLIDE COPY

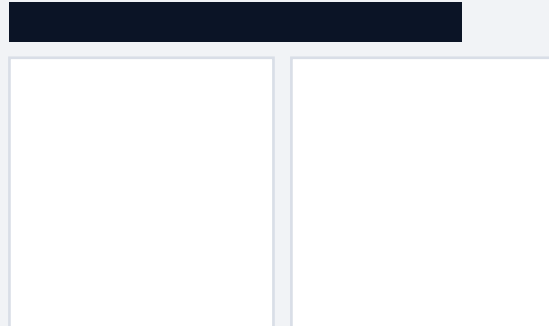
Two parallel decompositions for two batch regimes

- Batch DP: speedup = $\min(B, N)$; no inter-node forward messages
- Token-row split: even/odd causal query rows across two boards
- Parity imbalance: 1.6% vs 3× for a contiguous split

COMPOSITION INSTRUCTION

Use two equal technical diagrams. Left: independent batch samples to boards. Right: an S×S causal attention triangle colored in alternating query rows, with bidirectional K/V exchange.

WIREFRAME ONLY



NO EXTERNAL ASSET

DATA / EVIDENCE

Token-row mode exchanges 131,200 B per board in each direction per forward; weights remain replicated.

MUST PRESERVE

K and V move between peers; queries and assigned output rows remain local. Do not call token-row split tensor parallelism.

SOURCE PATH

docs/report/index.html
docs/MULTI_ESP32_DESIGN.md

Report both honest baselines

Judge takeaway: Eight nodes scale ideally for sufficient batch, but the WiFi-capable single node is slower than the best untiled node.

EXACT ON-SLIDE COPY

8.00× node scaling; 3.78× compute-wall gain

- 8.00× vs one tiled WiFi node
- 3.78× compute wall vs best untiled single board
- Case 2 token-row split: 1.56× on two boards · host transfer excluded

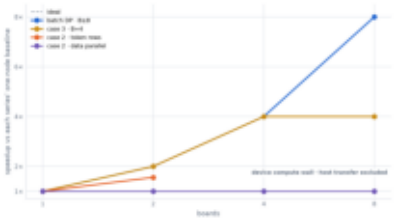
213 / 213 WiFi forwards pass · cases 1-5

COMPOSITION INSTRUCTION

Use the measured scaling plot. Put the two baseline definitions in separate callouts of equal visual weight.



ASSET PREVIEW · fig11 cluster scaling.svg



DATA / EVIDENCE

Per-device MFU: best opt23 42.2%; token-row split 32.9%; tiled WiFi replicas 19.9%.

MUST PRESERVE

Case 2 data parallel stays at 1× because B=1. Case 3 saturates at 4× because B=4.

SOURCE PATH

docs/report/index.html
benchmarks/README.md

Give the jury the complete benchmark accounting

Judge takeaway: Ten official shapes have physical-device measurements; status and numerical gate are separate facts.

EXACT ON-SLIDE COPY

10 of 14 official shapes ran on hardware

- Complete-case seconds; host transfer excluded
- One C3 · two C3 · eight-node WiFi
- Measured / partial / not run shown explicitly

COMPOSITION INSTRUCTION

Build a compact 14-row table from benchmark_cases.csv. Use seven columns: case, shape, 1 C3, 2 C3, 8 C3 WiFi, status, gate. Highlight case 2 only.

WIREFRAME ONLY



NO EXTERNAL ASSET

DATA / EVIDENCE

Footnotes: * B=1 activates one DP node. ** B=4 activates four nodes. Case 6 is host-only/partial; 8,13,14 are not run.

MUST PRESERVE

Do not put dashes in a green PASS style. Do not merge status and gate. Keep all time values in seconds.

SOURCE PATH

figure_assets/benchmark_cases.csv
docs/report/index.html

Turn non-running cases into a precise systems boundary

Judge takeaway: Each failure mode has a different remedy; hiding them would weaken the engineering story.

EXACT ON-SLIDE COPY

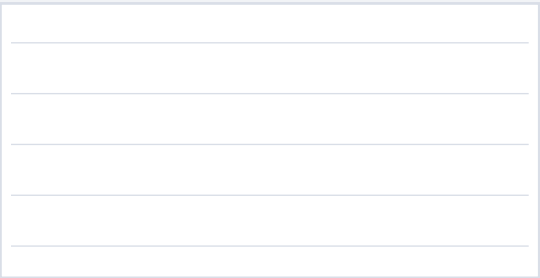
Four cases do not run—and the arithmetic says why

- Case 6: time scale · host gate only
- Cases 8/13: 50.4 MB int16 weights / 524 KB activation exceed flash / SRAM
- Case 14: 26.2 GB full-batch KV state

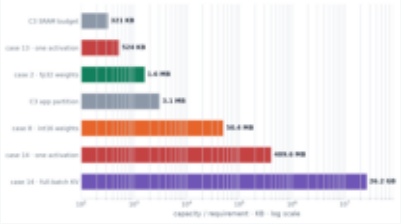
COMPOSITION INSTRUCTION

Use four horizontal rows, not four decorative cards. Each row must be formatted as requirement → board limit → next architectural step.

WIREFRAME ONLY



ASSET PREVIEW · fig02 memory scale.svg



DATA / EVIDENCE

Next steps: case 8 weight sharding; case 13 activation tiling + online attention; case 14 requires a different machine/system scale.

MUST PRESERVE

No device runtime for cases 6/8/13/14. Do not present projected board counts as achieved scaling.

SOURCE PATH

docs/report/index.html
benchmarks/README.md

Show the profiler as part of the contribution

Judge takeaway: tinyprof made cycle, call-tree, and static-memory bottlenecks reproducible on the C3.

EXACT ON-SLIDE COPY

We built the measurement tool this port was missing

- 6.25 ns cycle-counter zones + nested-call correction
- ELF census found a missing 32 KB scratch buffer
- [INSERT DEVICE TIMING COMPARISON AFTER CAPTURE]

COMPOSITION INSTRUCTION

Use a four-stage pipeline: cycle zones → call tree → ELF census → report. Reserve the right third for photo_04_profiler_bench.

WIREFRAME ONLY

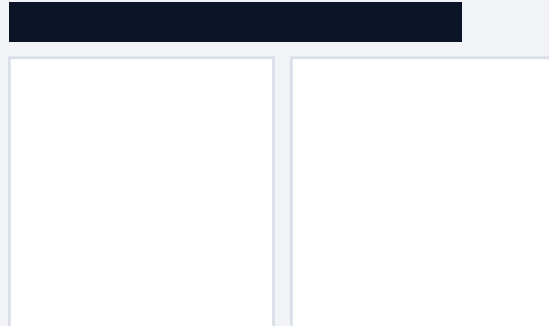


PHOTO · photo_04_profiler_bench

DATA / EVIDENCE

Build/host findings: 768 KiB weight traffic and 9,216 B LayerNorm reads. Device captures are still marked outstanding in the tinyprof log; leave the result panel as a memo until captured.

MUST PRESERVE

Before submission, add the exact AI tool model/version and the prompt → host gate → device profile → accept/revert workflow.

SOURCE PATH

benchmarks/case-02/optimisation/
esp32-baseline/optimisations/
25_tinyprof.md

Close on contributions and the honest frontier

Judge takeaway: The result is a complete numerical, kernel, memory, and distributed-systems co-design for a severe MCU target.

EXACT ON-SLIDE COPY

A tiny machine forced better systems thinking

- 21.1× one-board case-2 speedup
- 1.276 s two-board token-row forward
- 10/14 official shapes measured · zero failing elements across reported gates

COMPOSITION INSTRUCTION

Use four large numeric outcomes, then one restrained line of next steps: online long-sequence attention · weight sharding · recover tiled-node MFU.

WIREFRAME ONLY



NO EXTERNAL ASSET

DATA / EVIDENCE

Technical contributions: fixed-point attention, integer residual dataflow, RV32 kernels, SRAM-aware WiFi execution, two parallel modes, and tinyprof.

MUST PRESERVE

Add an explicit scope footer: device compute wall; host transfer excluded; seeded random weights; four cases without device runs.

SOURCE PATH

docs/report/index.html
benchmarks/README.md